

Data Science Project: Sales Performance Analysis & Prediction

Project Objective

Analyze historical sales data to discover trends, visualize performance, and predict future sales using machine learning.

Dataset

Synthetic retail sales dataset containing order date, region, product category, sales, and profit.

Tools & Technologies

Python (Pandas, NumPy, Matplotlib, Seaborn), Scikit-learn, Jupyter Notebook

Project Workflow

1. Data Cleaning & Preprocessing
2. Exploratory Data Analysis (EDA)
3. Data Visualization
4. Feature Engineering
5. Machine Learning Model (Linear Regression)
6. Model Evaluation

Key Insights

- Seasonal sales trends observed
- Certain regions consistently outperform others
- Machine learning model predicts sales with good accuracy

GitHub Repository Structure

```
project/  
datasets/  
notebooks/  
models/  
README.md
```

Conclusion

This project demonstrates end-to-end data science skills including data analysis, visualization, and predictive modeling.